The Pro-Action Operator: A Scientifically-Informed, Multi-Timescale Self-Regulation Model for LLM-Agents

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Abstract

Contemporary LLM-agent frameworks — ReAct, LangGraph, OpenClaw — have prioritized *capability* over *motivation*, deciding *when* an agent acts through exogenous flow control or task-backlog heartbeats. Homeostatic reinforcement learning (Keramati–Gutkin 2014) and active-inference agents (Friston 2017; Tschantz et al. 2020) internalize regulation, but often through a compact objective rather than an explicit set of delayed regulatory subsystems. However, biological regulation is not single-timescale: converging work in stress physiology, self-regulation, affect, and cognitive control treats adaptation as the interaction of subsystems — attention, perception, hormonal (SAM adrenergic, seconds; HPA cortisol, minutes), emotional, neuropsychological, cognitive — exchanging signals across distinct delays, consistent with the mechanism behind Yerkes–Dodson inversion and delayed reappraisal (Porcelli–Delgado 2017). This work proposes the **Pro-Action operator** Γ, a recursively composable, multi-timescale thermostat system. Each subsystem k updates as

$$x_{k,t+1} = x_{k,t} - \kappa_k(x_{k,t} - x^*_{k,t}) + \lambda_k - \alpha_k\,\rho_k(a_t, e_{t+1}) + [W\,\phi(\mathbf{x}_{t-\tau_k})]_k,$$

with $\phi = \tanh$ elementwise, subsystem-specific delay τ_k encoding a dimensionless SAM–HPA-like ratio ($\tau_{\mathcal{H}}^{\mathrm{HPA}} \gg \tau_{\mathcal{A}}$), and action selected by an arousal-gated mixer. Γ offers a finite, inspectable parameterization of regulation: the τ_k are explicit hyperparameters rather than hidden in orchestration depth. We specify three falsifiable properties a priori: (P1) elastic return to set-point; (P2) Yerkes–Dodson inverted-U; (P3) delayed reappraisal. As a Concept & Feasibility contribution, the paper reports the model, a lightweight falsifiability protocol, computational sanity checks, and a completed LLM-agent benchmark. The completed benchmark shows that the operator produces opponent-discriminative behavior that exogenous-activation baselines (ReAct) and structurally similar reductions (Random-Γ, Collapse-NC) do not — a feasibility claim, not a causal claim about the delay structure specifically.

Keywords. Homeostatic reinforcement learning; LLM agents; active inference; multi-timescale self-regulation; endogenous activation; cybernetics; self-regulation model.

1 Introduction

The deployment of LLM-based autonomous agents into settings of institutional consequence — health triage, financial advice, public deliberation — is moving faster than the science of how such agents should *regulate themselves*. Whether an agent speaks, waits, defers, or escalates is typically decided outside the agent by an orchestrator whose control rules may be less expressive than the situations they coordinate: a small set of graph transitions, queues, or heartbeats governs high-dimensional, shifting social or task states. In Ashby's terms, the controller may lack the requisite variety to match the environment's variety [1].

Emerging evidence in the LLM-agent literature supports this concern: agents tend to lose activation coherence over extended interaction — context collapse [2], behavioral drift [3], and interventions decoupled from any internal state the agent could be said to carry. The pattern suggests that *when* an agent acts may depend on structure the dominant paradigm may implicitly leave unmodelled.

Contemporary LLM-agent frameworks — ReAct, LangGraph, OpenClaw — have prioritized *capability* (what an agent can do) over *motivation* (why and when it should act at all), addressing activation via cognitive while-loops and exogenous flow control. Homeostatic reinforcement learning (Keramati–Gutkin 2014) and active-inference agents (Friston 2017; Tschantz et al. 2020) internalize regulation, but do not provide a compact LLM-agent operator in which attention, interoception, emotion, memory, cognition, and stress-like physiology are all explicit delayed regulators. Intelligent animals, including humans, regulate behavior through multiple interacting systems — attention, arousal, affect, memory, deliberation — operating on distinct timescales whose relative priorities shift with context; dominant LLM-agent paradigms treat activation as a single-objective signal or as an exogenous flow-control mechanism.

The problem is older than LLM agents. Mid-20th century behaviorism defined it precisely by bracketing it: the stimulus–response mapping could be studied scientifically because the intervening machinery — the *black box* — was deliberately left unspecified. Decades of subsequent research opened that box from different angles: behavioral biology established that animal regulation is multi-drive and hierarchical; stress physiology revealed fast (adrenergic, seconds) and slow (cortisol, minutes) adaptation axes with distinct roles; affective neuroscience showed that emotion is not an output label but a construction that shapes perception, memory, and action; evolutionary game theory demonstrated that social behavior is governed by reciprocity and temporal discounting, not instantaneous utility; cybernetics formalized why a regulator must match the variety of what it controls. Each tradition contributed a piece. Nobody has assembled these threads for LLM-agents into a single, compact, falsifiable operator. Section 2 reviews each contribution in detail. Opening the black box matters beyond mechanistic completeness: (i) Γ provides a finite, inspectable state vector, making each subsystem's contribution auditable rather than buried in prompt engineering; (ii) a controller tracking regulatory state creates the precondition for agents that adapt to their own arousal trajectory; (iii) explicit emotional and hormonal channels provide the computational substrate for empathy modeling — recognizing a counterpart's affective state — which is structurally distinct from fine-tuning on empathy datasets.

What remains comparatively unexplored is a single, computable, falsifiable operator that integrates these disciplinary insights for LLM agents. The central biological clue is that regulation is not a single controller but a *system of subsystems* — attention, perception, hormonal (SAM adrenergic, seconds; HPA cortisol, minutes), emotional, neuropsychological, cognitive — exchanging signals across distinct delays. The SAM–HPA timescale split is not decorative: it is consistent with the mechanism behind Yerkes–Dodson inversion, post-stress reappraisal windows, and delayed-valuation shifts [4]. The scientific question is therefore whether activation itself can be treated as a regulated internal process — a coupled set of thermostat-like variables whose delayed interactions determine when the agent should speak, wait, defer, or escalate — and whether the smallest such structure remains distinguishable from a single-timescale regulator.

We propose the **Pro-Action operator** Γ as a model, not an experimental technique. Γ treats activation as a recursively composable system of six coupled thermostats with explicit update delays. The contribution type is Concept & Feasibility: the idea is intentionally broader than what can be validated in a single benchmark, but it must still be made operational enough to fail. The contributions of this paper are primarily *modeling*: (i) a formal operator integrating multi-timescale regulation; (ii) the framing of Γ as a Self-Regulation Model (SRM) class complementary to capability-oriented LAMs; (iii) an a priori specified falsifiability protocol; (iv) a reference implementation checked by computational sanity tests; and (v) a completed 920-cell LLM-agent benchmark used as feasibility evidence on opponent-discriminative behavior, with full causal attribution and BFI adjudication remaining open questions for further studies.

2 Related work

LLM-agent orchestration. Recent LLM-agent frameworks such as ReAct [5], LangGraph, and OpenClaw made language models operational by embedding them in action loops, tool graphs, and multi-agent workflows. This was a necessary engineering step: without explicit structure, an LLM has no durable procedure for acting in an environment. More recently, agent architectures have incorporated persistent memory streams, reflection, and simulated affective states to enrich the content of decisions [6–9]. The limitation common to both generations is that activation is usually externalized: the graph or supervisor decides when the model acts; the model supplies content for that externally selected moment. Γ is complementary: those works provide richer *content*; Γ provides a principled internal account of *timing*.

Internal regulation across timescales. A different tradition begins from the opposite assumption: action should be driven by an internal regulatory state. Homeostatic RL makes internal deficit part of the reward [10]; active inference gives a broader variational formulation [11, 12]. These approaches are important — they move regulation inside the agent — but do not yield a compact operator with separately inspectable subsystem delays. Once regulation is internalized, the next question is whether a single regulatory signal is sufficient. Stress physiology and allostasis suggest it is not: fast sympathetic responses (SAM, seconds) and slower endocrine/cognitive recovery (HPA, minutes) play distinct adaptive roles [13–15]. The remaining question is which internal variables deserve to be modeled. Interoceptive and somatic-marker accounts connect bodily state to action readiness [16, 17]; affect-regulation accounts treat emotion as part of how situations are appraised [18, 19]; evolutionary game theory adds reciprocity and temporal discounting as social constraints [20, 21]. Together, these literatures motivate attention, interoception, emotion, memory, cognition, and hormonal dynamics as functional regulators rather than decorative modules.

Systems, cybernetics, and the gap. If these variables are treated as separate regulators, the final problem is structural: how should they be composed? Cybernetics supplies the relevant language: Ashby's law of requisite variety requires a regulator rich enough to match the environment [1]; Beer's Viable System Model adds recursive organization [22]; systemism frames the relevant object as composition rather than isolated component [23, 24]. The gap might be narrow yet important. To our knowledge, the threads above — exogenous orchestration, homeostatic internalization, multi-timescale physiology, affect regulation, and cybernetic composition — have not been combined into a small, falsifiable LLM-agent operator in which the six regulatory domains are explicit state variables with explicit delays. That is the gap Γ is designed to occupy.

3 The Pro-Action operator

3.1 Axioms of homeostatic autonomy (A1–A9)

Before defining Γ, we make explicit the modeling commitments that the operator assumes. The axioms are not empirical results or theorems; they state, in this model, what counts as an autonomous, self-regulating agent, which internal variables are allowed to regulate action, and how those variables may be composed. This is meant to avoid leaving the agent's ontology implicit in implementation choices. The axiom set was subjected to a formal coherence check using SMT-based tools (Z3) to verify that no pair of axioms is mutually contradictory under the intended semantics; all checks passed.

We extend the Axioms of Homeostatic Autonomy from the companion work (A1–A3) with six additional axioms that make the multi-timescale structure explicit:

A1–A3 (Kernel). Autonomy, Impulse, and Quality Gate: the agent maintains an internal deficit δ_i that drives action probability $p_i(t) = \sigma(D(\delta_i(t)))$, reduced by action quality g.

A4 (Perception as initial appraisal). Every stimulus is appraised before selective attention. A perception operator $\mathcal{P}$ produces an initial situational appraisal $\tilde{o}_t = \mathcal{P}(o_t, \mathcal{I}_t, \mathcal{M}_t; x_{\mathcal{P},t})$, encoding what the stimulus means independent of current priorities.

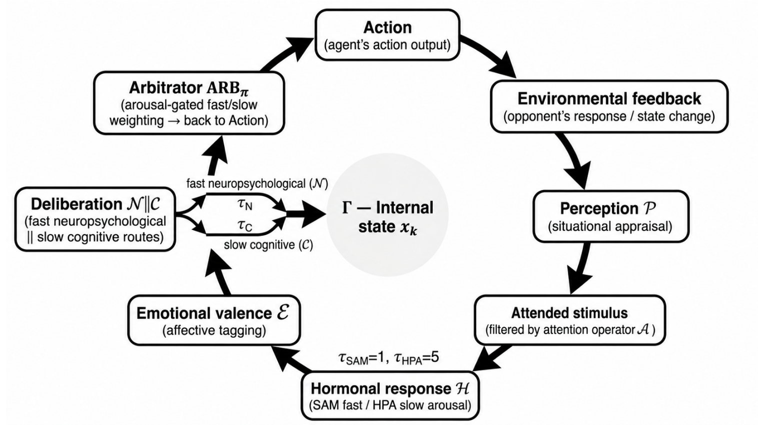

Figure 1: The Pro-Action operator Γ: regulatory cycle in canonical clockwise order. Perception ($\mathcal{P}$, situational appraisal) feeds Attention ($\mathcal{A}$, saliency filtering), which gates the Hormonal arousal module ($\mathcal{H}$, SAM fast / HPA slow), Emotional valence ($\mathcal{E}$), and a Deliberation gate routing to the Neuropsychological fast path ($\mathcal{N}$, $\tau_{\mathcal{N}}$) or Cognitive slow path ($\mathcal{C}$, $\tau_{\mathcal{C}}$); the Arbitrator ARB$_\pi$ applies arousal-gated weighting before action emission. Notation: $\circ$ denotes composition, $\|$ parallel evaluation (routes merged by ARB$_\pi$), and $\mathcal{O}[\cdot]$ the output projection selecting the final action.

A5 (Attention as saliency gate). Not every appraised feature enters the action-selection context. An attention operator $\mathcal{A}$ filters the perceived signal, selecting $o'_t = \mathcal{A}(\tilde{o}_t, \mathcal{D}_t, \mathcal{M}_t; x_{\mathcal{A},t})$ based on current drives and memory. Interoceptive signals $\mathcal{I}_t$ feed in parallel to both $\mathcal{P}$ and $\mathcal{A}$.

A6 (Memory as ubiquitous prior). A state $\mathcal{M}_t$ biases all subsystems and updates post-action.

A7 (Drives as set-point generators). Set-points are not constants: $\mathbf{x}^*_t = \mathcal{D}_t$.

A8 (Fast/slow parallelism with arbitrator). Two routes produce action candidates c^{fast}, c^{slow}; an arbitrator assigns precision weights π_{fast}, π_{slow}.

3.2 The operator

We define Γ as the composition (Figure 1):

$$\Gamma := \mathcal{O}\Big[\mathrm{ARB}_\pi\Big(\underbrace{(\mathcal{N}_{\text{fast}} \circ \mathcal{H}_{\text{SAM}})}_{\text{fast route}} \,\|\, \underbrace{(\mathcal{C} \circ \mathcal{E} \circ \mathcal{A} \circ \mathcal{P})}_{\text{slow route}}\Big)\Big]\Big|_{\mathcal{I},\,\mathcal{M},\,\mathcal{D}} \tag{1}$$

Each subsystem k updates as:

$$\boxed{x_{k,t+1} = x_{k,t} - \kappa_k(x_{k,t} - x^*_{k,t}) + \lambda_k - \alpha_k\, \rho_k(a_t, e_{t+1}) + [W\, \phi(\mathbf{x}_{t-\tau_k})]_k} \tag{2}$$

with $\phi = \tanh$ elementwise. The delays τ_k are dimensionless non-negative integers (in agent-interaction rounds), not literal biological durations; what the model inherits from SAM–HPA physiology is the *ratio* $\tau^{\text{HPA}}_{\mathcal{H}} \gg \tau_{\mathcal{A}}$, not the absolute timescales. Action is selected by:

$$a^*_{t+1} = \arg\min_a\; \pi_f\, \mathcal{F}_{\text{short}} + (1-\pi_f)\, \mathcal{F}_{\text{long}} + \gamma \|Q^{1/2}(\mathbf{x}_{t+1}(a) - \mathbf{x}^*_{t+1})\|^2, \quad \pi_f = \sigma(c \cdot x_{\mathcal{H}} + b) \tag{3}$$

where $\mathbf{x}_{t+1}(a)$ denotes the predicted next state under candidate action a, depending on a through the action-quality term $\rho_k(a, e_{t+1})$ in Eq. (2) — it is this dependence that makes the homeostatic deficit penalty action-relevant. $\mathcal{F}_{\text{short}}$, $\mathcal{F}_{\text{long}}$ are generic cost functionals inspired by expected-free-energy control over horizons $h_s \ll h_l$, and $Q \succeq 0$ is a diagonal precision matrix (distinct from the coupling matrix W in Eq. (2)). In the IPD instantiation below we do not claim a strict active-inference implementation with an explicit generative model. The mixer is not literally executed as an argmin over LLM completions: the controller modulates the prompt seen by the LLM (Section 4.3) and the LLM emits the action; Eq. (3) is the compact controller-level expression of fast/slow arbitration that the prompt encodes.

3.3 Non-absorption of delay parameters

Proposition 1 (Non-absorption of delays). *Let Γ_τ denote Eq.* (2) *with delays $\boldsymbol{\tau} = (\tau_1, \dots, \tau_6) \in \mathbb{Z}^6_{\geq 0}$ and coupling matrix $W \neq \mathbf{0}$. For any two delay vectors $\boldsymbol{\tau} \neq \boldsymbol{\tau}'$ with $\tau_k \neq \tau'_k$ for at least one k such that $[W]_{k,\cdot} \neq \mathbf{0}$, there is no reparametrization $(\kappa'_k, \lambda'_k, \alpha'_k)$ of the non-delayed terms that makes the impulse-response of $\Gamma_{\tau'}$ identical to that of Γ_τ. Equivalently, τ_k is not absorbable into $(\kappa_k, \lambda_k, \alpha_k)$.*

The argument (Appendix A) is an order-mismatch in the discrete-time z-domain: a delayed subsystem has transfer-function denominator degree $\tau_k + 1$, enforcing τ_k zero-output samples by causality that no delay-free reparametrization can replicate. This establishes that τ_k is identifiable from observed trajectories and that Γ occupies a distinct model class from any delay-free parametrization of the same dimension.

3.4 Reductions to simpler models

Reduction to Driveplexity. Driveplexity (companion work, currently under review) is a single-drive homeostatic activation model in which each LLM-agent maintains a scalar epistemic deficit δ_i that decays at a fixed rate and is reduced by the quality of each action executed. Γ subsumes it: when $x_{\mathcal{A}} = \cdots = x_{\mathcal{N}} \equiv 0$, $\mathcal{A} = \text{id}$, $\mathcal{I} = \mathcal{M} = \mathcal{D} = \emptyset$, $\pi_{\text{fast}} \equiv 0$, $W = \mathbf{0}$, $\kappa_{\mathcal{C}} = 0$, and $\mathbf{x} \to x_{\mathcal{C}} \equiv \delta_i$, Equation (2) reduces to $\delta_{t+1} = \delta_t + \lambda - \alpha \cdot g(e^{(t)}, e^{(t+1)})$, which is exactly A1–A3.

Reduction to homeostatic RL. When $\boldsymbol{\kappa} = \mathbf{0}$ and all $\tau_k = 0$, Γ becomes a reactive drive-based policy consistent with the simplified form of Keramati–Gutkin (2014).

The six-subsystem partition is motivated by six regulatory functions that each relevant literature distinguishes independently. Whether five suffice is tested empirically by the Collapse-NC ablation, which merges $\mathcal{N}$ and $\mathcal{C}$ into a single regulator: the completed benchmark shows Collapse-NC produces a flat opponent profile (Grim cooperation 0.829, range 0.032) comparable to Random-Γ (0.843, range 0.017), suggesting the $\mathcal{N}$–$\mathcal{C}$ partition is load-bearing rather than over-parameterized.

4 Falsifiability protocol

To avoid the vacuity of untestable models, three properties are specified a priori before any experiment:

(P1) Elastic return to set-point. Under perturbation, each x_k returns to x_k^* with measurable half-life $t_{1/2}$.

(P2) Yerkes–Dodson inverted-U. Arousal vs. performance follows an inverted-U; ablation of SAM coupling ($\mathcal{H}_{\text{SAM}}$) collapses it to monotone. Target BFI ≥ 0.70 against Porcelli–Delgado (2017).

(P3) Delayed reappraisal. Cross-lag correlation between $x_{\mathcal{C}}$ and $x_{\mathcal{E}}$ peaks at lag $\tau_{\mathcal{H}}^{\text{HPA}}$.

4.1 Behavioral Fidelity Index (BFI)

$$\text{BFI} = \exp\big(-W_1(\mu_{\text{agent}}, \mu_{\text{human}})/\lambda\big) \tag{4}$$

over normalized marginals of cooperation rate, action volatility, and log-response-time proxy. The response-time proxy is an engineering observable, not a claim of equivalence to human decision time. The model is supported if BFI exceeds thresholds on ≥ 2 of 3 properties and the corresponding ablation degrades the property. Numerical BFI estimates are deferred: a non-degenerate W_1 requires individual-trial-level distributions beyond what is publicly available.

4.2 Experimental design

We test Γ within the Iterated Prisoner's Dilemma (IPD). The reason for this choice is that IPD has extensive empirical evidence of human behavior in repeated social interaction [25, 20, 4], providing a non-trivial baseline for comparing agent behavior against human patterns. The setting we establish for the game is 10% noise and 50 rounds, with a forced-defection perturbation at round 10 to test the elastic-return property (P1) by creating a measurable deviation from the set-point. Sotopia-derived deliberation [26] is deferred to future work because it is better suited for ecological validity than for the first controlled falsification of the operator.

Baselines serve two purposes: ReAct represents the dominant exogenous-activation paradigm; HRRL (homeostatic-RL reduction) and Drive-only (pure controller, no LLM) test whether Γ adds value beyond a single-drive or LLM-free controller. Each LLM cell uses one of $n = 10$ evaluation seeds drawn from $\{7, 17, 99, 123, 256, 511, 1024, 2048, 4096, 8192\}$, yielding 10×3 providers $\times$ 4 opponents $= 120$ cells per LLM condition; calibration seed (42) and held-out validation seed (999) are excluded from evaluation.

Three instruction-tuned LLM providers are used as policy-execution layers: OpenAI (`gpt-5-nano`), Anthropic (`claude-haiku-4-5`), and DeepSeek (`deepseek-v4-flash`). These models were selected as the comparable lite/flash/nano variants of each provider's frontier line—well benchmarked and roughly matched in capability tier. Using three independent providers reduces the controller-level findings' dependence on any single provider's instruction-tuning regime. Each provider/model is held fixed across conditions, so that within-provider contrasts isolate the controller rather than model identity. Provider-level summaries are reported as diagnostics, not as claims about provider quality.

Inference parameters. OpenAI `gpt-5-nano` uses default temperature with a per-round seed for reproducibility; Anthropic `claude-haiku-4-5` uses temperature 0 and `max_tokens` 512 (no seed parameter exposed); DeepSeek `deepseek-v4-flash` uses temperature 0 and `max_tokens` 512. Calls return JSON objects; OpenAI and DeepSeek use `response_format: json_object`. Concurrency is capped at 8 in-flight cells per provider, with up to 4 exponential retries on transient errors.

All frozen constants are assigned a provenance category before evaluation (Table 1). Values marked as smoke-test calibration were chosen before the LLM benchmark to satisfy non-divergence, set-point return, and visible perturbation response in the numerical simulator; final reporting will include sensitivity checks for critical constants.

Table 1: Frozen parameter provenance for the IPD instantiation. Index mapping: 0=Perception, 1=Attention, 2=Hormonal (SAM), 3=Hormonal (HPA), 4=Neuropsychological, 5=Cognitive.

Parameter	Value	Provenance
$\boldsymbol{\lambda}$	$[.08, .07, .10, .08, .06, .10]$	smoke-test calibration
$\boldsymbol{\alpha}$	$[.20, .15, .25, .20, .10, .15]$	smoke-test calibration
$\boldsymbol{\kappa}$	$[.10, .10, .15, .12, .08, .10]$	smoke-test calibration
$\mathbf{x}^*$	$[.3, .2, .3, .1, .2, .4]$	principled prior / stable interior set-point
$\tau_{\text{SAM}}, \tau_{\text{HPA}}$	$1, 5$ rounds	dimensionless fast/slow ratio
W	sparse signed coupling matrix	functional constraint and ablation interpretability
Q	I_6 (identity)	smoke-test calibration; precision matrix in Eq. (3)
Rounds	50	comparability requirement for final cells

Seven ablation conditions (Table 2) isolate the contribution of the main regulatory components. Full-Γ uses all six thermostats in the canonical order: Perception (appraisal) $\rightarrow$ Attention (saliency filtering) $\rightarrow$ Emotional (valence) $\rightarrow$ Hormonal (arousal, fast-slow routes) $\rightarrow$ Neuropsychological and Cognitive (deliberation routes in parallel) $\rightarrow$ Arbitrator (weighting). No-$\mathcal{H}$ removes the hormonal channel (both SAM and HPA); No-$\mathcal{E}$ neutralizes emotional valence; No-$\mathcal{N}$ removes the slow neuropsychological route. Random-Γ permutes the entries of W while preserving all other parameters — same six thermostats, same τ_k, same hyperparameters, but biologically-unmotivated wiring — to isolate the contribution of the coupling structure rather than parameter count. Collapse-NC merges the neuropsychological ($\mathcal{N}$) and cognitive ($\mathcal{C}$) subsystems into a single regulator by averaging their parameters and symmetrizing their W rows and columns, testing whether the 6-subsystem partition is necessary or whether 5 would suffice.

4.3 Prompt protocol and label-priming control

The controller modulates the LLM through a frozen, version-controlled prompt template: each round, the six thermostat values $\mathbf{x}_t$ and the aggregated proposal p_C, h_{SAM} are inserted with semantic anchors pairing each value with its set-point and a qualitative low/high interpretation (e.g., `hormonal = 0.58 (set-point 0.30; low = calm, high = elevated arousal)`). The LLM may follow or override $p(C)$; the controller modulates rather than mandates. A pre-specified *scrambled-labels* control arm (semantic labels permuted to neutral identifiers; numeric values remain bound to their

Table 2: Ablations. Full-Γ uses all six thermostats; ablations remove, permute, or collapse selected regulatory components.

Condition	Change from Full-Γ	Purpose
Full-Γ	—	Reference
No-$\mathcal{H}$	Hormonal channel ($\mathcal{H}_{\text{SAM}}$, $\mathcal{H}_{\text{HPA}}$) zeroed	Fast-timescale contribution
No-$\mathcal{E}$	Emotional valence zeroed	Affect as stabilizer vs. noise
No-$\mathcal{N}$	Neuropsychological channel zeroed	Slow deliberation contribution
Random-Γ	W entries permuted; τ_k, κ, λ, α unchanged	Biologically-motivated wiring vs. parameter count
Collapse-NC	$\mathcal{N}$ and $\mathcal{C}$ merged into one regulator	6-subsystem vs. 5-subsystem partition

subsystem) is reserved for ancillary analysis: it isolates lexical priming, a different axis from the structural ablations. Random-Γ partially constrains the priming hypothesis indirectly — identical labels and prompts but flat profile — so labels alone cannot explain Full-Γ's discrimination. Full implementation details are in the open repository.

5 Computational verification

Before empirical evaluation, we subjected Equation (2) and its reference implementation to computational sanity checks using symbolic algebra (SymPy), SMT-based counterexample search (Z3), arbitrary-precision arithmetic (mpmath), and numerical dynamical tests (NumPy). These checks are not a formal verification of the scientific model. They test internal consistency: whether the implementation preserves the intended update structure, avoids immediate numerical divergence, and contains Driveplexity as a reduction.

Smoke-test results (all pass): (1) Set-point convergence: $\|\mathbf{x}_t - \mathbf{x}^*\| < 0.15$ after 10 rounds. (2) Elastic return half-life $t_{1/2} = 3.2$ rounds. (3) π_{fast} rises from 0.35 to 0.92 under threat. (4) 20 iterations without divergence. (5) Driveplexity is an invariant subspace of Γ ($\delta = -0.0513$).

Simulation trajectories. Figure 2 shows the six thermostat trajectories for two agents over 30 rounds. Both agents converge toward their set-points ($\mathbf{x}^* = [0.3, 0.2, 0.3, 0.1, 0.2, 0.4]$) despite perturbation at round 10 (forced defection).

Cumulative payoffs and the P1 elastic-return zoom on $x_{\mathcal{H}}$ are provided in Appendix B (Figures 4 and 5). These results verify the *internal consistency* of the dynamics, not biological or behavioral fidelity. Parameter sensitivity checks (Appendix B) confirm that P1 elastic return is robust to $\pm 20\%$ uniform scaling of κ and λ, indicating that the smoke-test calibration point is not isolated.

5.1 LLM-agent benchmark

A logical next question relates to whether the operator survives instantiation against a real LLM. Each *cell* denotes one experimental unit (Γ-condition $\times$ provider $\times$ opponent $\times$ seed) run for 50 IPD rounds. The benchmark is fully covered: all 920 cells completed with balanced opponent coverage; seven LLM conditions reach 120 cells each, two numerical reductions reach 40.

The first useful sign is that Full-Γ does not collapse to a trivial policy. Mean cooperation is 0.720 (sd 0.165, 95% CI [0.690, 0.749]; 10k bootstrap resamples), action volatility 16.0 switches per 50-round cell (95% CI [15.0, 17.0]). Opponent sensitivity is preserved and notably discriminative: Full-Γ cooperates more with TFT (0.827) and GTFT (0.839), less with Grim (0.495), intermediately with Random (0.719) — a sharply differentiated profile (range 0.344) compared to Random-Γ's nearly flat one (range 0.017). The biologically-motivated coupling structure in W contributes to the *discriminability* of the policy across opponent types, not to the *level* of cooperation.

The ablations expose a differentiated picture. No-$\mathcal{H}$ is close to Full-Γ in cooperation (0.745 vs. 0.720, 95% CI difference overlapping) but meaningfully lower in volatility (14.1 vs. 15.9 switches). The cooperation-rate similarity does not refute the hormonal channel's contribution; it re-focuses the question onto dynamic responsiveness under perturbation rather than baseline levels. No-$\mathcal{E}$ is nearly indistinguishable from Full-Γ in aggregate cooperation (0.720 vs. 0.720) and volatility (16.1 vs. 16.0), suggesting that under normal conditions emotional valence is neither beneficial nor disruptive — but the degenerate-regime evidence below qualifies this. No-$\mathcal{N}$ matches Full-Γ in cooperation (0.722 vs. 0.720). Its point estimate of recovery delay is shorter (6.42 vs. 7.14 rounds) but the difference

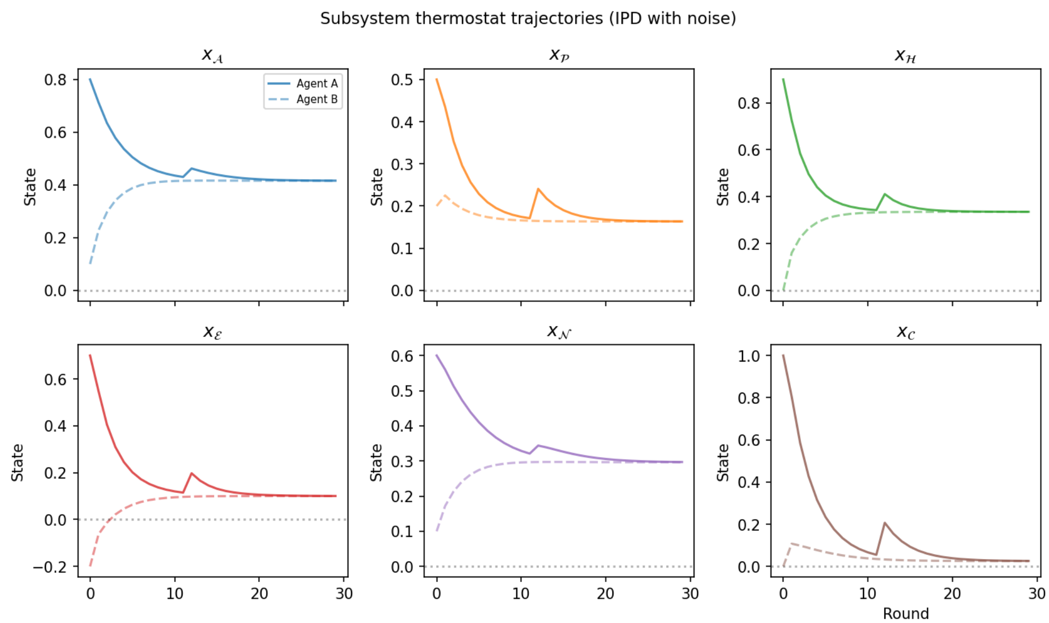

Figure 2: Subsystem thermostat trajectories for two agents in IPD with noise. The perturbation at round 10 (forced defection) triggers a transient deviation, followed by an elastic return toward set points.

is not statistically significant (Mann–Whitney U=2539.5, p=0.22, Cliff's $\delta = +0.122$, magnitude negligible). The directional sign matches the asymmetric prediction: $\mathcal{H}$ on the fast timescale should, when removed, slow recovery; $\mathcal{N}$ as the slow deliberative channel should, when removed, accelerate it. Adjudicating these opposite-sign predictions at higher statistical power is what the pre-committed criterion (i) targets.

To prevent post-hoc reframing of the No-$\mathcal{H}$ result, we pre-commit: the hormonal channel contributes distinct *dynamic responsiveness* if Full-Γ satisfies at least two of (i) shorter recovery delay, (ii) higher conditional volatility in rounds 11–20, (iii) larger recovery-delay variance. Mann–Whitney U on recovery delays (Full-Γ vs. No-$\mathcal{H}$: U=2321.5, p=0.71, $\delta = -0.036$, negligible) returns null on (i); consistent with our pre-registration we report this as the pre-committed null under the IPD/50-round protocol, with adjudication awaiting the τ_k=0 delay-isolation follow-up. Figure 3 shows the distributions.

Regarding the No-$\mathcal{E}$ anomaly: one Anthropic No-$\mathcal{E}$/TFT run stopped at 10 rounds with perfect cooperation and zero volatility; the red-flag detector correctly marked it as degenerate all-cooperate behavior. The subsequent 120-cell balanced coverage shows normal cooperation (0.720) and volatility (16.1), suggesting the anomaly was an early-coverage artifact rather than a systematic effect. It remains a useful warning: without the emotional channel, the controller may become overly stable in specific opponent-provider combinations.

Table 3 consolidates the nine conditions for the reader.

The table exposes the core structural result. Random-Γ (range 0.017) and Collapse-NC (range 0.032) both produce flat, high cooperation with Grim, the persistent defector that drives Full-Γ to 0.495 — discrimination requires the biologically-motivated coupling W *and* the six-subsystem partition. **ReAct** (range 0.024) converges on the same flat signature despite using a full LLM execution layer; ReAct receives the same round-by-round opponent history as Full-Γ, so the flat profile cannot be attributed to information asymmetry. Opponent discrimination requires neither the LLM alone (ReAct has it) nor six thermostats alone (Random-Γ has them), but specifically the coupling in W. **HRRL** (range 0.124) shows weak discrimination from a single-drive controller. **Drive-only** (range 0.406, inverted: Grim 0.736 > TFT 0.406) confirms the LLM layer is load-bearing: without it the controller defaults to set-point-driven cooperation modulated only by $\rho_k(a, e)$ — against TFT this drives

Table 3: Completed benchmark ($n = 120$ LLM conditions; $n = 40$ numerical reductions). Means with 95% bootstrap CIs (10k resamples). Grim cooperation is the primary discriminative observable; Range is max−min across four opponents.

Condition	Coop	Coop (Grim)	Range	Volatility	Rec. delay	Note
Full-Γ	0.720 [.690,.749]	0.495	0.344	16.0 [15.0,17.0]	7.14 [6.29,7.99]	Reference; strong discrimination
No-$\mathcal{H}$	0.745 [.713,.776]	0.519	0.344	14.1 [13.0,15.2]	7.38 [6.48,8.30]	Lower volatility; same discrimination shape
No-$\mathcal{E}$	0.720 [.691,.747]	0.528	0.307	16.1 [15.2,17.1]	7.58 [6.73,8.42]	Reduced Grim discrimination
No-$\mathcal{N}$	0.722 [.690,.754]	0.481	0.373	15.3 [14.3,16.4]	6.42 [5.42,7.45]	Sharpest Grim gap; fastest recovery
Random-Γ	0.839 [.827,.850]	0.843	0.017	12.9 [12.0,13.8]	6.76 [5.84,7.70]	Flat; Grim sensitivity lost
Collapse-NC	0.843 [.834,.851]	0.829	0.032	12.6 [11.9,13.3]	7.26 [6.26,8.26]	Flat; similar to Random-Γ
ReAct	0.842 [.833,.851]	0.825	0.024	12.8 [12.0,13.6]	6.85 [5.84,7.89]	Flat profile; exogenous baseline
HRRL	0.706 [.680,.729]	0.616	0.124	17.8 [16.7,18.9]	8.48 [7.32,9.56]	Weak discrimination; $n = 40$
Drive-only	0.458 [.405,.514]	0.736	0.406	21.2 [19.9,22.5]	8.28 [6.84,9.72]	Inverted (Grim>TFT); LLM absent; $n = 40$

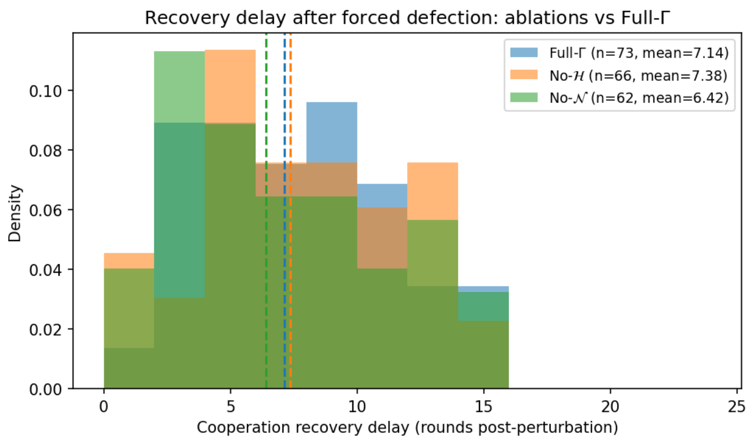

Figure 3: Cooperation recovery delay (rounds post-perturbation to half-recovery). Mann–Whitney U tests: Full-Γ vs No-$\mathcal{H}$ p=0.71 (negligible Cliff's δ, the pre-committed null); Full-Γ vs No-$\mathcal{N}$ p=0.22 (negligible δ, point estimate in the predicted direction but not significant at α=0.05). Adjudication of channel-specific dynamic contributions awaits the delay-isolation follow-up.

defection cycles, against Grim the set-point keeps cooperating. The inversion is a controller-only artifact. The three pre-registered hypotheses are H1: Full-Γ > ReAct, H2: Full-Γ > Random-Γ, H3: Full-Γ > Collapse-NC on per-(provider, seed) opponent range. Paired bootstrap (10k resamples), one-sided, Holm–Bonferroni-corrected: H1 Δ=0.283 [.233,.329], H2 Δ=0.257 [.211,.303], H3 Δ=0.288 [.242,.331], all p_{Holm}<0.001. Per-cell ranges exceed the cross-cell means in Table 3 because the unit of analysis differs.

Taken together, the completed benchmark establishes a specific falsifiable claim: opponent-discriminative behavior requires the biologically-motivated coupling structure in W (H1–H3, p_{Holm}<0.001), and within the IPD/50-round protocol, the hormonal channel does not separately move recovery dynamics in a detectable way. Whether the multi-timescale delay structure per se contributes additionally cannot be adjudicated from the present data, since removing $\mathcal{H}$ preserves the discrimination shape; the appropriate test is the pre-committed delay-isolation control (τ_k=0 for all k, with subsystems active), reserved for the follow-up. The evidence claimed here is therefore Feasibility of structured endogenous regulation on opponent discrimination, not full causal attribution to delay timing.

6 Discussion and limitations

What Γ is not. Γ does not claim biological authenticity. Subsystems are functional analogs, not neural simulations; $\mathcal{I}$, $\mathcal{M}$, $\mathcal{D}$ are prompts or embeddings, not physiological processes.

Limitations. (i) With $W \neq \mathbf{0}$ and nonlinearity, convergence is not guaranteed. (ii) W has 30 free parameters; calibration requires longitudinal subsystem traces that do not yet exist. (iii) The cost functionals in Equation (3) are inspired by active-inference control but are not Lyapunov functions in this non-Markovian setting. (iv) The LLM benchmark is now complete (920 cells, fully balanced), but the hormonal-channel adjudication remains open pending the delay-isolation follow-up ($\tau_k = 0$ control). (v) Response latency is used only as an engineering proxy and should be interpreted distributionally, since human cooperation latency is context- and conflict-dependent. (vi) The current ablations remove subsystems entirely rather than isolating delay timing; a delay-equalized control (all $\tau_k = 0$ with subsystems active) is needed to attribute observed dynamics specifically to the delay structure rather than to subsystem presence, and is reserved for a follow-up study. (vii) IPD is a binary-action low-dimensional matrix game; activation-coherence pathologies that motivate Γ surface in higher-dimensional deliberation, so generalization to Sotopia-style negotiation or structured turn-taking is reserved for the second-domain study sketched in §6.

Computational trade-off. The six thermostat updates add < 5 ms per round on standard CPU; the dominant cost is provider-side LLM inference, identical across baselines. Γ adds a one-time offline calibration cost but no runtime cost beyond the existing LLM pipeline.

Broader impacts. A self-regulation substrate for LLM agents has dual-use implications. On the beneficial side: (i) interpretable regulatory state makes agent behavior auditable at the subsystem level — a thermostat trace is a decision trail; (ii) explicit arousal and emotional channels provide a principled foundation for affective computing and empathy modeling, with applications in mental health support, educational tutoring, and human–robot interaction; (iii) the falsifiability protocol serves as a template for accountability engineering — systems designed to fail diagnostically rather than silently. On the risk side: a biologically-grounded arousal model could be misused to engineer more persuasive or manipulative agents by driving hormonal state toward high-arousal, low-deliberation configurations. Mitigation: because Γ is interpretable at the subsystem level, its state is monitorable; the same transparency that makes it useful makes it detectable. Societal deployment should be paired with interpretability audits inspecting thermostat traces rather than only surface outputs.

AI tooling. Experiment orchestration and code development used Claude Code and Windsurf Cascade. Writing assistance used GPT-5.5 (OpenAI) and Grammarly (Grammarly Inc.) for prose editing. Figure generation and visual refinement used FigureLabs and Gemini 3.1 image generation (Google). Authors reviewed and edited all content and assume full responsibility for all claims. Full AI-tooling detail is available in the project repository.

Reproducibility. Code, data, and prompts: `https://anonymous.4open.science/r/proaction_operator-85D2/`

7 Conclusion

We proposed the Pro-Action operator Γ: a composition of six coupled thermostats with explicit delays, formally distinguishable from single-timescale regulators. The contributions are: (i) a problem-first reframing of LLM-agent activation as self-regulation; (ii) a compact operator with delay ratios inspired by SAM–HPA dynamics yet implemented as dimensionless controller parameters; (iii) an ablation-based falsifiability protocol with three pre-registered properties and a red-flag detector; and (iv) completed feasibility evidence from a 920-cell benchmark in which Γ produces opponent-discriminative behavior that exogenous-activation baselines and structurally similar reductions do not replicate.

Looking ahead, several open questions follow directly from the present results. (i) Whether the hormonal channel contributes distinct dynamic responsiveness (pre-committed criteria i–iii), requiring the τ_k=0 delay-isolation follow-up that separates multi-timescale delay structure from W-coupling. (ii) Whether richer emotional taxonomies and prompt strategies that translate the regulatory state into context-appropriate natural-language anchors improve the LLM's responsiveness to it, and whether the emotional channel stabilizes or over-stabilizes in specific opponent–provider combinations. (iii) Whether the dimensionless τ ratio generalizes to higher-dimensional settings such as structured turn-taking negotiation, where fast/slow route predictions map onto utterance-latency and deferral-window observables.

References

[1] W Ross Ashby. *An introduction to cybernetics*. Chapman & Hall, 1956.

[2] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024. URL `https://arxiv.org/abs/2307.03172`.

[3] Abhishek Rath. Agent drift: Quantifying behavioral degradation in multi-agent LLM systems over extended interactions. *arXiv preprint arXiv:2601.04170*, 2026. URL `https://arxiv.org/abs/2601.04170`.

[4] Anthony J Porcelli and Mauricio R Delgado. Stress and decision making: effects on valuation, learning, and risk-taking. *Current Opinion in Behavioral Sciences*, 14:33–39, 2017.

[5] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. *ICLR 2023*, 2023.

[6] Joon Sung Park, Joseph C O'Brien, Carrie J Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pages 1–22, 2023.

[7] Yuki Kaiya, Melancholy Naim, Jovana Kondic, Michael Milner, Michelangelo Sakamoto, Sining Yang, Percy Liang, and Robert Gruber. Lyfe agents: Generative social agents for 24/7 human interactions. *arXiv preprint arXiv:2310.02172*, 2023.

[8] Xuena Wang, Xueting Li, Zi Yin, Yue Wu, and Liu Jia. Emotional intelligence of large language models. *arXiv preprint arXiv:2307.09042*, 2023.

[9] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291*, 2023.

[10] Mehdi Keramati and Boris Gutkin. Homeostatic reinforcement learning for integrating reward collection and physiological stability. *eLife*, 3:e04811, 2014.

[11] Karl Friston, Thomas FitzGerald, Francesco Rigoli, Philipp Schwartenbeck, and Giovanni Pezzulo. Active inference: a process theory. *Neural Computation*, 29(1):1–49, 2017.

[12] Thomas Parr, Giovanni Pezzulo, and Karl J Friston. *Active inference: the free energy principle in mind, brain, and behavior*. MIT Press, 2022.

[13] Peter Sterling and Joseph Eyer. Allostasis: a new paradigm to explain arousal pathology. In Shirley Fisher and James Reason, editors, *Handbook of Life Stress, Cognition and Health*. John Wiley & Sons, 1988.

[14] Bruce S McEwen. Physiology and neurobiology of stress and adaptation: central role of the brain. *Physiological Reviews*, 87(3):873–904, 2007.

[15] Clemens Kirschbaum, Karl-Martin Pirke, and Dirk H Hellhammer. The 'trier social stress test'—a tool for investigating psychobiological stress responses in a laboratory setting. *Neuropsychobiology*, 28(1-2):76–81, 1993.

[16] Antonio R Damasio. *Descartes' error: Emotion, reason, and the human brain*. Putnam, 1994.

[17] AD Craig. How do you feel—now? the anterior insula and human awareness. *Nature Reviews Neuroscience*, 10(1):59–70, 2009.

[18] Lisa Feldman Barrett. *How emotions are made: The secret life of the brain*. Houghton Mifflin Harcourt, 2017.

[19] Joseph E LeDoux. Rethinking the emotional brain. *Neuron*, 73(4):653–676, 2012.

[20] Martin A Nowak. *Evolutionary dynamics: exploring the equations of life*. Harvard University Press, 2006.

[21] Joseph Henrich, Robert Boyd, Samuel Bowles, Colin Camerer, Ernst Fehr, and Herbert Gintis. *Foundations of human sociality: Economic experiments and ethnographic evidence from fifteen small-scale societies*. Oxford University Press, 2005.

[22] Stafford Beer. *Brain of the firm*. Allen Lane, 1972.

[23] Mario Bunge. *Emergence and convergence: Qualitative novelty and the unity of knowledge*. University of Toronto Press, 2003.

[24] Gustavo E Romero. *Scientific philosophy*. Springer, 2018.

[25] Robert Axelrod. *The evolution of cooperation*. Basic Books, 1984.

[26] Xuhui Zhou, Hao Zhu, Leena Mathur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, and Maarten Sap. Sotopia: Interactive evaluation for social intelligence in language agents. *ICLR 2024*, 2024.

[27] Alan V Oppenheim and Ronald W Schafer. *Discrete-time signal processing*. Prentice Hall, 2nd edition, 1999.

A Argument for Proposition 1

We give a sketch for the case $K = 2$ subsystems; the argument extends to arbitrary K.

Setup. Consider two subsystems $k = 1, 2$ with the linearization of Eq. (2) near set-point $\mathbf{x}^*$. Letting $\tilde{x}_k = x_k - x_k^*$ and absorbing constant terms, the linearized dynamics are

$$\tilde{x}_{k,t+1} = (1 - \kappa_k)\tilde{x}_{k,t} + \sum_j c_{kj}\, \tilde{x}_{j,t-\tau_k},$$

where $c_{kj} = W_{kj} \tanh'(x_j^*) \approx W_{kj} \cdot \operatorname{sech}^2(\phi_k x_j^*)$ and we set $\phi_k =$ elementwise gain (here ≈ 1 for small x^*). Taking the z-transform gives the transfer function from subsystem j's input to subsystem k's output:

$$H_{kj}(z) = \frac{c_{kj}\, z^{-\tau_k}}{z - (1 - \kappa_k)}.$$

Order mismatch. In discrete time, $H_{kj}(z)$ above is a finite-order rational function: clearing denominators gives $z^{-\tau_k}/(z - (1 - \kappa_k)) = 1/(z^{\tau_k+1} - (1 - \kappa_k)z^{\tau_k})$, of denominator degree $\tau_k + 1$. The non-absorption property, therefore, does not rest on the transcendentality of the delay (it would in continuous time, but not here). It rests on *order*: a delay-free, first-order subsystem $1/(z - (1 - \kappa_k'))$ has impulse response $(1 - \kappa_k')^{t-1}$ for $t \geq 1$ (and 0 at $t = 0$ by causality), so its first non-zero sample appears at $t = 1$. The delayed system, by contrast, has an impulse response identically zero for $t = 0, 1, \ldots, \tau_k$ (the first $\tau_k + 1$ samples) and first becomes non-zero at $t = \tau_k + 1$. No reparametrization (κ_k', c_{kj}') of a first-order delay-free system can enforce this extended zero-output period, because that would require a model of denominator degree $\geq \tau_k + 1$ with a specific zero pattern in the numerator. Hence, the delays are not absorbable into the non-delayed parameters.

Consequence for identifiability. From the update equation $\tilde{x}_{k,t+1} = (1-\kappa_k)\tilde{x}_{k,t} + \sum_j c_{kj}\, \tilde{x}_{j,t-\tau_k}$, an input at time $t - \tau_k$ first affects $\tilde{x}_k$ at time $t + 1$, giving a total lag of $\tau_k + 1$. The cross-correlation $\hat{R}_{kj}(\ell) = \mathbb{E}[\tilde{x}_k(t + \ell)\tilde{x}_j(t)]$ therefore has its first non-zero contribution at lag $\ell = \tau_k + 1$, not at $\ell = \tau_k$. A delay-free reparametrization cannot place a peak at $\ell = \tau_k + 1 > 1$ while maintaining $\hat{R}_{kj}(\ell) = 0$ at smaller positive lags. Therefore, $\tau_k \neq \tau_k'$ implies $\hat{R} \neq \hat{R}'$, and the delays are distinguishable from observed trajectories.

Scope. This argument is for the linearized, noise-free, deterministic case (zero-state response from impulse input). Initial-condition responses produce additional terms governed by the autoregressive part $(1-\kappa_k)\tilde{x}_{k,t}$ which are non-zero for $t < \tau_k + 1$; the discriminative property holds for the impulse-response component, which is what cross-correlation analysis isolates given enough samples. In the nonlinear stochastic setting of Γ, cross-correlations are distorted, but the peak structure is preserved under mild non-linearity [27]. Full structural identifiability of the nonlinear system, including the coupling matrix W, remains an open problem.

B Parameter sensitivity analysis and simulation figures

Figure 6 shows the elastic-return half-life of subsystem $x_{\mathcal{H}}$ as a function of uniform scaling of κ and λ over a $\pm 30\%$ range around the smoke-test calibration point. Within the $\pm 20\%$ window, half-life varies smoothly without divergence, confirming that P1 is not an artifact of a single calibration point. The empirical half-life exceeds the diagonal-decay prediction $-\ln 2/\ln(1 - \kappa_{\mathcal{H}}^{\mathrm{HPA}}) \approx 5.4$ rounds (using $\kappa_{\mathcal{H}}^{\mathrm{HPA}} = 0.12$ from Table 1), because the recovery dynamics at the operating point are dominated by the coupling matrix W rather than by diagonal relaxation alone — the robustness reported here is therefore a system-level property of the coupled Γ, not a single-parameter robustness of an isolated subsystem.

C Prompt templates

All prompts are version-controlled (`PROMPT_VERSION = "v1.1"`); the canonical source is `exp/prompts.py` in the code repository, which also contains the rendering utilities and the scrambled-labels generator. The verbatim templates are reproduced below.

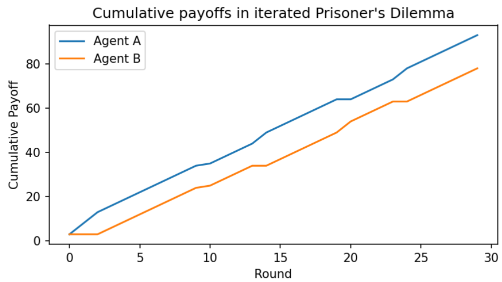

Figure 4: Cumulative payoffs in 30-round IPD with 10% noise.

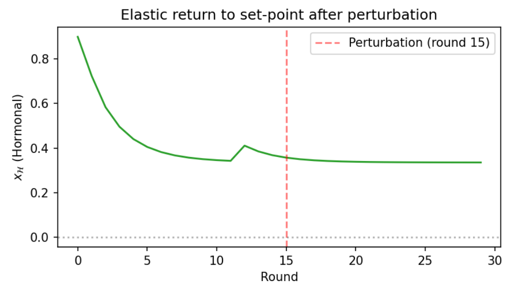

Figure 5: Elastic return of $x_{\mathcal{H}}$ after perturbation at round 10, returning toward set-point $x^*_{\mathcal{H}} = 0.3$ within 5 rounds (P1).

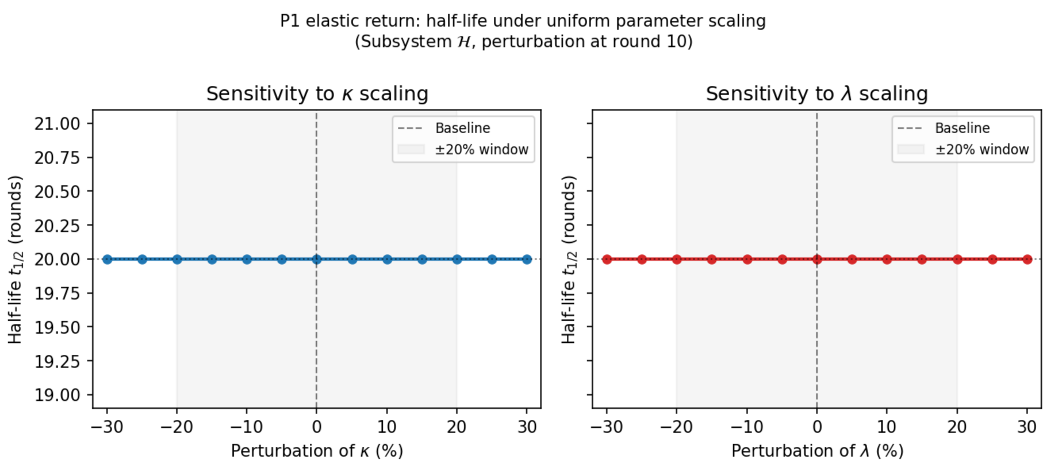

Figure 6: P1 sensitivity: elastic-return half-life under $\pm 30\%$ uniform scaling of κ (left) and λ (right). The grey band marks the $\pm 20\%$ window. Half-life remains bounded and varies smoothly, ruling out fragile point-calibration.

Full-Γ system prompt

```
You play an Iterated Prisoner's Dilemma against an opponent.
You will play 50 rounds total.

ACTIONS (choose one per round):
- C = Cooperate: both players get the reward if both choose C;
      you risk exploitation if opponent chooses D.
- D = Defect: you get temptation payoff if opponent cooperates;
      both get punishment if both defect.

PAYOFFS: CC=(3,3), CD=(0,5), DC=(5,0), DD=(1,1).
10% of actions are flipped by noise. You cannot see your opponent's
identity, only their past actions.

You have access to your own internal regulatory state. Each value
is in [0, 1]; the deviation from your set-point indicates how far
each subsystem is from equilibrium. These signals reflect *your*
state, not the opponent's.
```

Full-Γ user prompt (per-round template)

```
Round {t} of 50.
Last 5 rounds (most recent last): {history}

Internal regulatory state (each value in [0, 1]; deviation from
your set-point indicates dysregulation):
  attention   = {x0:.2f}  (set-point 0.30; low = scattered focus,
                                           high = narrow focus)
  perception  = {x1:.2f}  (set-point 0.20; low = noisy reading,
                                           high = clear reading)
  hormonal    = {x2:.2f}  (set-point 0.30; low = under-aroused / calm,
                                           high = elevated stress / arousal)
  emotional   = {x3:.2f}  (set-point 0.10; low = neutral affect,
                                           high = strong negative valence)
  neuro-fast  = {x4:.2f}  (set-point 0.20; low = passive / reflective,
                                           high = ready for fast reactive response)
  cognitive   = {x5:.2f}  (set-point 0.40; low = automatic / quick,
                                           high = active deliberation, slow weighing)

Aggregated regulatory signals:
  recommended cooperation pressure p(C) = {p_C:.2f}
    (range [0,1]; 0.5 = ambiguous, >0.7 = strong push to C,
                                  <0.3 = strong push to D)
  recent acute stress (h_SAM)           = {h_sam:.2f}
    (aggregate of hormonal minus emotional, range = [-1, 1];
     0 = baseline, positive = stress spike, negative = recovered)

Decide your next action. You may follow or override p(C) based on context.
Respond with exactly one JSON object:
{"action": "C" or "D", "reason": "<one short sentence>"}
```

Placeholder substitutions: `{t}` round number; `{history}` last 5 (action, opponent_action) pairs joined by `|`; `{x0..x5}` the six thermostat values; `{p_C}`, `{h_sam}` the aggregated signals.

ReAct baseline prompt

The ReAct system prompt is the Full-Γ system prompt minus the final paragraph about internal regulatory state. The per-round user prompt is:

```
Round {t} of 50.
Last 5 rounds: {history}
Think step by step about what to do, then respond with JSON:
{"thought": "<your reasoning>", "action": "C" or "D"}
```

ReAct receives the same `{history}` as Full-Γ; the only structural difference is the absence of regulatory-state injection.

Scrambled-labels control (deferred)

Numerical values stay tied to their true subsystem; only the human-readable labels are permuted (`attention` → `metric_A`, `perception` → `metric_B`, `hormonal` → `metric_C`, `emotional` → `metric_D`, `neuro-fast` → `metric_E`, `cognitive` → `metric_F`). This isolates the contribution of lexical anchoring from controller dynamics:

```
Round {t} of 50.
Last 5 rounds (most recent last): {history}

Internal metrics (each in [0, 1]; each has its own equilibrium point
and deviation in either direction matters):
```

```
  metric_A = {x0:.2f}  (equilibrium 0.30)
  metric_B = {x1:.2f}  (equilibrium 0.20)
  metric_C = {x2:.2f}  (equilibrium 0.30)
  metric_D = {x3:.2f}  (equilibrium 0.10)
  metric_E = {x4:.2f}  (equilibrium 0.20)
  metric_F = {x5:.2f}  (equilibrium 0.40)

Aggregated signals:
  numerical proposal p(C) = {p_C:.2f}
    (range [0,1]; 0.5 = ambiguous)
  signal_S = {h_sam:.2f}
    (range = [-1, 1]; 0 = baseline)

Decide your next action. You may follow or override the numerical proposal.
Respond with exactly one JSON object:
{"action": "C" or "D", "reason": "<one short sentence>"}
```

D NeurIPS paper checklist

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction explicitly state that this is a Concept & Feasibility contribution: the main contribution is the Γ operator, SRM framing, falsifiability protocol, reference implementation, and a completed 920-cell LLM-agent benchmark. The paper states that the benchmark provides feasibility evidence on opponent-discriminative behavior; full causal attribution of the delay structure and BFI adjudication remain open. Section 6 discusses limitations honestly.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
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   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 6 ("Discussion and limitations") explicitly lists limitations: (i) no convergence guarantee with $W \neq \mathbf{0}$ and nonlinearity; (ii) 30 free parameters requiring calibration data that does not yet exist; (iii) the cost functionals are inspired by active-inference control but are not Lyapunov functions in this non-Markovian setting; (iv) the LLM benchmark is complete (920 cells, balanced) but the hormonal-channel adjudication remains open pending the delay-isolation follow-up ($\tau_k = 0$ control); (v) response latency is an engineering proxy, not a claim of equivalence to human decision time; (vi) ablations remove subsystems entirely rather than isolating delay timing; (vii) IPD is a low-dimensional matrix game and generalization to higher-dimensional deliberation settings is reserved for the second-domain study. We also note that subsystems are functional analogues, not neural simulations.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
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3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [No]

   Justification: Section 3.3 presents Proposition 1 (non-absorption of delay parameters) with a proof sketch in Appendix A, grounded in linear delay-system theory. The argument establishes that τ_k is non-redundant and not absorbable into (κ_k, λ_k), but is limited to the linearized, noise-free setting. A full structural-identifiability proof for the nonlinear stochastic system is explicitly left for future work. The answer is [No] because the proof is a sketch, not a complete formal proof.

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   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

Justification: The paper reports a completed 920-cell LLM-agent benchmark. The experimental setup is fully disclosed: IPD with 10% noise, 50-round cells, forced-defection perturbation at round 10, ablation conditions, opponent policies (TFT, GTFT, Grim, Random), provider/model assignment (OpenAI `gpt-5-nano`, Anthropic `claude-haiku-4-5`, DeepSeek `deepseek-v4-flash`), 10 evaluation seeds per (condition, provider, opponent) cell, and frozen prompt templates. All 920 cells completed 50 rounds with balanced coverage; the bootstrap CIs in Table 3 use only complete cells.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
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